Information Theory and Statistical Learning

From Shannon to PAC learning, from entropy to generalization

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Abstract

We will cover information theory, estimation, learning theory, kernels, neural network theory, and online learning.

Contents

1 PART I — INFORMATION THEORY

1.1 1. Entropy and Mutual Information

- Shannon entropy; joint, conditional, mutual information
- Relative entropy (KL divergence); properties; chain rules
- Differential entropy; Gaussian channels; entropy of mixtures
- Data processing inequality; Fano's inequality

1.2 2. Lossless Source Coding

- Asymptotic equipartition property (AEP); typical sequences
- Shannon's source coding theorem; prefix codes; Huffman coding
- Arithmetic coding; Lempel-Ziv universal compression
- Kolmogorov complexity (brief); connections to learning

1.3 3. Channel Coding

- Channel capacity; Shannon's noisy-channel theorem; proof via typicality
- Gaussian channel; water-filling; multi-antenna (MIMO) brief
- Linear codes; Reed-Solomon (from finite fields, Course 2); LDPC
- Polar codes and capacity-achieving constructions (brief)

1.4 4. Information-Theoretic Inequalities

- Log-sum inequality; convexity of KL divergence
- Fisher information; Cramér-Rao bound; van Trees inequality
- Entropy power inequality; de Bruijn's identity
- Connections to Course 1 (Sections 4, 28) and Course 5

1.5 5. Information Theory and Statistics

- Hypothesis testing; Neyman-Pearson lemma; error exponents
- Chernoff bound; Stein's lemma; Hoeffding bound
- Method of types; empirical entropy; connection to large deviations
- Universal hypothesis testing; I-projection

1.6 6. Information Theory and Machine Learning

- Information bottleneck; rate-distortion perspective on representations
- MI estimation with neural networks (MINE, NWJ bounds)
- Differential privacy: formal definition; mechanisms; composition
- Rate-distortion theory and representation learning; connections to VAEs

2 PART II — STATISTICAL ESTIMATION THEORY

2.1 7. Classical Estimation

- Maximum likelihood; consistency; asymptotic normality; efficiency
- Cramér-Rao bound; Fisher information matrix; Bhattacharyya bound
- M-estimators; robustness; breakdown point; influence function
- Bayesian estimation; conjugate priors; posterior; minimax Bayes

2.2 8. Exponential Families and Sufficient Statistics

- Natural parameters; log-partition function; moment parameterization
- Sufficient statistics; Fisher-Neyman factorization; completeness
- Rao-Blackwell theorem; Lehmann-Scheffé theorem
- Connections to information geometry (Course 7, Section 6)

2.3 9. Minimax Theory and Optimal Rates

- Minimax risk; parameter space complexity; Le Cam's two-point method
- Assouad's lemma; Fano's inequality applied to estimation
- Minimax rates for nonparametric regression and density estimation
- Adaptive estimation: Lepski's method; oracle inequalities; wavelet thresholding

3 PART III — STATISTICAL LEARNING THEORY

3.1 10. PAC Learning Framework

- PAC model; sample complexity; ε-δ learnability; realizable and agnostic
- VC dimension; growth function; Sauer-Shelah lemma
- Fundamental theorem of statistical learning: VC characterizes PAC learnability
- Examples: halfspaces, decision trees, neural networks

3.2 11. Rademacher Complexity and Uniform Convergence

- Rademacher complexity; symmetrization lemma
- Covering numbers; metric entropy; Dudley's chaining integral
- Generalization bounds via Rademacher complexity; data-dependent bounds
- PAC-Bayes bounds; KL-regularized objectives; connections to Bayesian methods

3.3 12. Stability, Compression, and Generalization

- Algorithmic stability: uniform stability; connection to generalization
- On-average stability; SGD is stable; stability of noisy algorithms
- Compression-based bounds; mutual information bounds
- Connections to regularization; data augmentation as stability

3.4 13. Kernel Methods and RKHS

- Reproducing kernel Hilbert spaces; Mercer's theorem; feature maps
- Kernel ridge regression; SVMs; representer theorem
- Spectral analysis of kernels; eigenvalue decay; approximation rates
- Random features; Nyström approximation; scalable kernel methods

3.5 14. High-Dimensional Statistics

- Curse of dimensionality; concentration of measure; geometric results
- Lasso: restricted eigenvalue condition; oracle inequalities; de-biasing
- Covariance estimation; sample covariance in high dimensions; shrinkage
- Connections to compressed sensing (Course 1, Section 33)

3.6 15. Neural Network Theory

- Overparameterization; benign overfitting; interpolation threshold
- Neural tangent kernel (NTK): linearization; kernel regime; lazy training
- Mean field theory of two-layer networks; Wasserstein gradient flow
- Double descent: risk curve; model and epoch double descent
- Implicit regularization of gradient descent; connections to sparsity

3.7 16. Online Learning and Bandits

- Online convex optimization; regret; minimax regret; expert advice
- Hedge (Multiplicative Weights) algorithm; regret bounds; connections to entropy
- Multi-armed bandits: UCB; Thompson sampling; lower bounds; Lai-Robbins
- Contextual bandits; linear bandits; connections to RL